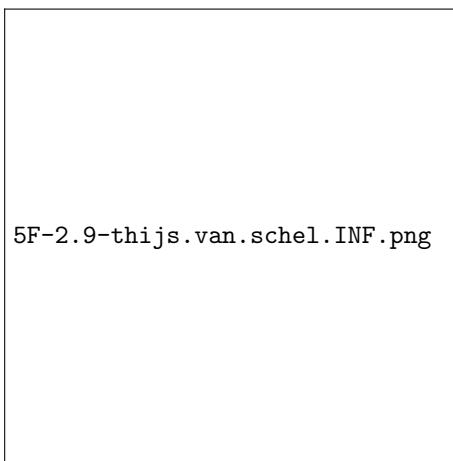


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$$\begin{aligned} A &= \frac{1}{2} \int_0^{2\pi} (r_1^2 - r_2^2) d\theta \\ \begin{cases} r_1 = 3 + \sin(\theta) \\ r_2 = 2 + \sin(\theta) \end{cases} \\ A &= \frac{1}{2} \int_0^{2\pi} ((3 + \sin(\theta))^2 - (2 + \sin(\theta))^2) d\theta \\ &= \frac{1}{2} \int_0^{2\pi} ((9 + 6\sin(\theta) + \sin^2(\theta)) - (4 + 4\sin(\theta) + \sin^2(\theta))) d\theta \\ &= \frac{1}{2} \int_0^{2\pi} (5 + 2\sin(\theta)) d\theta \\ &= \frac{1}{2} [5\theta - 2\cos(\theta)]_0^{2\pi} \\ &= \frac{1}{2} ((10\pi - 2\cos(2\pi)) - (0 - 2\cos(0))) \\ &= \frac{1}{2} ((10\pi - 2) - (-2)) \\ &= \frac{1}{2}(10\pi) \\ A &= 5\pi \end{aligned}$$

Tekening:



References